

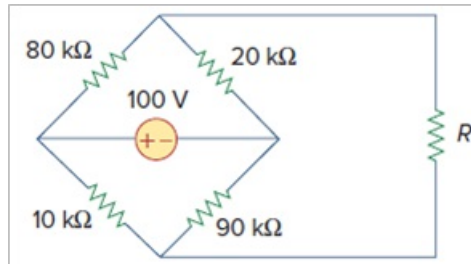
Problem

The variable resistor R in Fig. 4.133 is adjusted until it absorbs the maximum power from the circuit.

(a) Calculate the value of R for maximum power.

(b) Determine the maximum power absorbed by R .

Figure 4.133:



Step-by-step solution

Step 1 of 7

Refer to Figure 4.133 in the text book.

(a)

Set the value of the source to zero to calculate the value of the Thevenin's resistance.

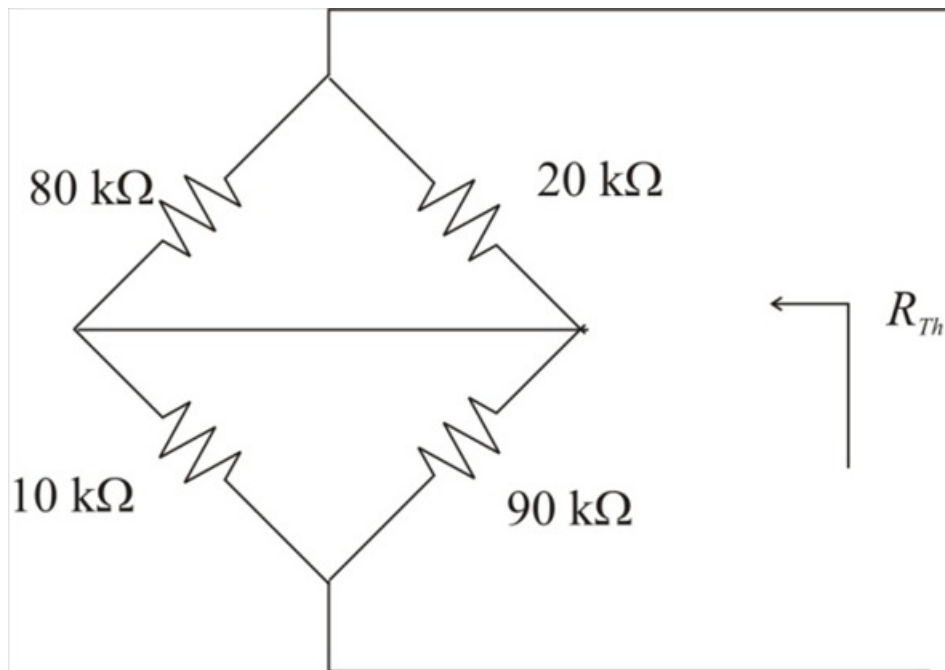


Figure 1

Step 2 of 7

Calculate the value of the Thevenin's resistance R_{Th} .

$$\begin{aligned} R_{Th} &= (80 \text{ k} \parallel 20 \text{ k}) + (10 \text{ k} \parallel 90 \text{ k}) \\ &= \frac{(80 \text{ k})(20 \text{ k})}{80 \text{ k} + 20 \text{ k}} + \frac{(10 \text{ k})(90 \text{ k})}{10 \text{ k} + 90 \text{ k}} \\ &= \frac{(80 \text{ k})(20 \text{ k})}{100 \text{ k}} + \frac{(10 \text{ k})(90 \text{ k})}{100 \text{ k}} \\ &= \frac{80 \text{ k}}{5} + \frac{90 \text{ k}}{10} \end{aligned}$$

$$\begin{aligned} R_{Th} &= 16 \text{ k} + 9 \text{ k} \\ &= 25 \text{ k}\Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance is $25 \text{ k}\Omega$.

For the load to absorb the maximum power, the value of the load resistance must be equal to the Thevenin's resistance. Therefore, the value of the load resistance R is $25 \text{ k}\Omega$.

Step 3 of 7

(b)

Disconnect the load resistance to calculate the value of the Thevenin's voltage.

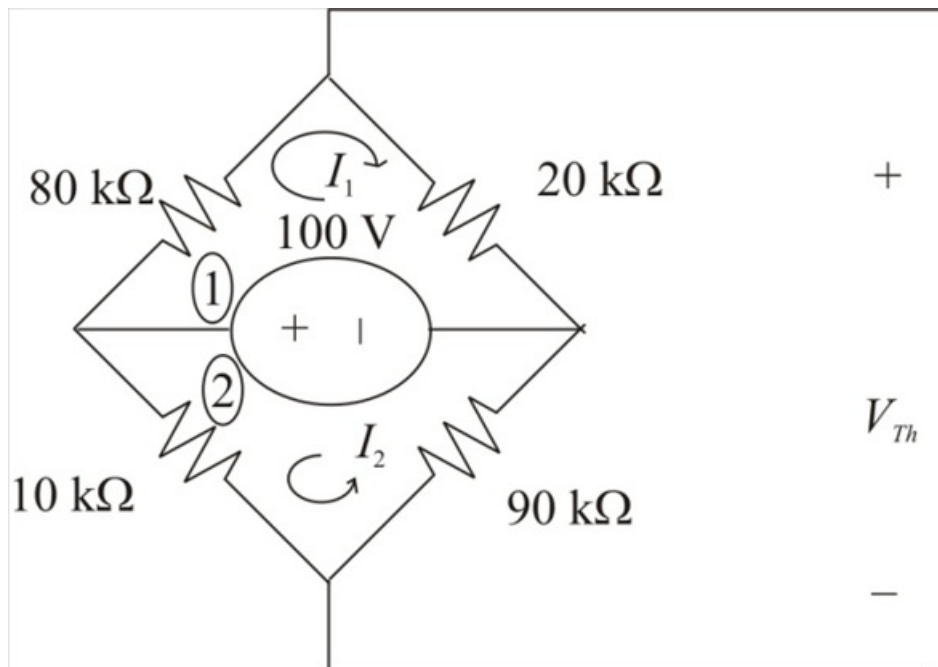


Figure 2

Step 4 of 7

Apply Kirchhoff's voltage law in loop-1.

$$-100 + I_1(80 \text{ k} + 20 \text{ k}) = 0$$

$$I_1(100 \text{ k}) = 100$$

$$I_1 = \frac{100}{100 \text{ k}}$$

$$I_1 = 1 \text{ mA}$$

Therefore, the value of the current I_1 is 1 mA .

Step 5 of 7

Apply Kirchhoff's voltage law in loop-2.

$$-100 + I_2(90 \text{ k} + 10 \text{ k}) = 0$$

$$I_2(100 \text{ k}) = 100$$

$$I_2 = \frac{100}{100 \text{ k}}$$

$$I_2 = 1 \text{ mA}$$

Therefore, the value of the current I_2 is 1 mA .

Step 6 of 7

Calculate the value of the Thevenin's voltage V_{Th} .

$$V_{\text{Th}} = I_1(20 \text{ k}) - I_2(90 \text{ k})$$

Substitute 1 mA for I_2 and I_1 .

$$\begin{aligned} V_{\text{Th}} &= (1 \text{ m})(20 \text{ k}) - (1 \text{ m})(90 \text{ k}) \\ &= 20 - 90 \\ &= -70 \text{ V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is -70 V .

Step 7 of 7

Calculate the maximum power absorbed by the load.

$$P_{\text{max}} = \frac{V_{\text{Th}}^2}{4R}$$

Substitute -70 V for V_{Th} and $25 \text{ k}\Omega$ for R .

$$\begin{aligned} P_{\text{max}} &= \frac{(-70)^2}{4(25 \text{ k})} \\ &= \frac{4900}{100 \text{ k}} \\ &= 49 \text{ mW} \end{aligned}$$

Therefore, the value of the maximum power absorbed by the load is 49 mW .